

L1. Content of recommendations:

Based on our testing results and expert feedback, we recommend the following improvements to enhance the device's safety, durability, and usability.

First, we should add safety guards around the gears and crank to prevent finger pinching, addressing a key safety concern raised by our reviewer.

Second, we recommend using more durable materials or reinforced wood components to reduce wear and tear and avoid issues like jamming or breaking during extended use.

Third, we suggest developing quantifiable testing procedures to better measure student engagement and motor skill improvement, which will help us refine the design based on objective data.

Finally, making the device modular or adjustable will allow customization to better meet the diverse needs of students with different abilities, increasing overall usability and effectiveness.

L2. Detailed plans for implementation of improvements:

To implement these recommendations, we will start by attaching the already bought silicone side protectors. For improving durability, we will research and source stronger wood types or composite materials. To quantify testing, we will develop clear observation protocols and simple metrics, such as counting grip adjustments or duration of engagement, and pilot these with small student groups for feedback. Lastly, we will redesign key components to be detachable or adjustable, such as interchangeable

handles or variable gear sizes, and create assembly instructions that allow easy customization depending on student needs. Each improvement will be tested iteratively, incorporating user and expert feedback to guide ongoing revisions.